Alpha Factor Mining Using Hierarchical Genetic Algorithms with Quality-Diversity Selection

Junyoung Jung

Abstract—Traditional alpha factors—cross-sectional signals that predict benchmark-relative excess returns for portfolio weighting—face mounting challenges in increasingly efficient markets. Individual alpha lifespans are shortening due to computational advances, while high-performing candidates often exhibit strong correlations that reduce portfolio-level diversification benefits. This paper addresses these limitations by combining hierarchical genetic algorithms with quality-diversity selection for alpha generation, then aggregating multiple factors into a "Mega Alpha" portfolio using deep neural networks. My framework starts with 3,000 depth-1 alpha expressions, then iteratively applies crossover, mutation, and PCA-grid selection to expand expressivity while controlling redundancy. Alpha quality is measured by Information Coefficient (IC). Diversity is enforced by selecting top-|IC| candidates per grid cell in a 2D PCA behavior space. Testing on S&P 100 data shows average IC improved from 0.001 to 0.004, with many factors exceeding the industry threshold $|\text{IC}| > 0.02$. The Mega Alpha portfolio achieved 60% cumulative returns and rolling Sharpe ratios between -1.0 to 4.2, but suffered 40% maximum drawdown due to long-only constraints. Results show that combining multiple alphas can overcome individual limitations while exposing vulnerabilities in down markets.

Index Terms—Alpha Factors, Evolutionary Algorithms, Quality–Diversity, PCA Grid, Deep Neural Networks, Portfolio Optimization, Explainable AI, Quantitative Finance.

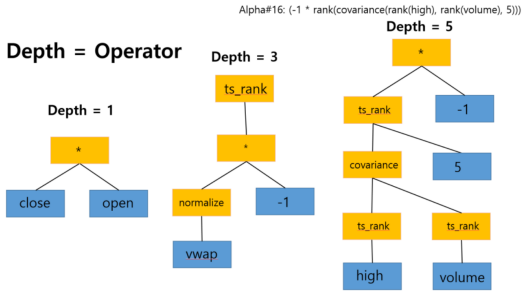


Fig. 1: A visual definition of hierarchical Alpha.

1 Introduction

In quantitative investing, the term *alpha* is used in two senses: (i) excess return versus a benchmark, and (ii) a *signal/strategy* intended to produce such excess return. This paper adopts the latter and restricts *alpha* to explicit, human-readable expressions constructed as hierarchical compositions of basic data fields (e.g., OHLCV, `adj_close`) and operators (arithmetic, transforms, time-series and cross-sectional). For a visual intuition of excess-return alpha, see [?].

Traditional discovery relies on human expertise and ad-hoc exploration. As market efficiency increases, finding persistent alpha requires systematic search methods that balance quality and diversity. This work contributes a hierarchical genetic algorithm with PCA-grid quality-diversity selection and a neural network aggregation method with SHAP explanations.

2 Related Work

Canonical resources include the definition and taxonomy of alpha [?] and the *101 Formulaic Alphas* [1]. Hierarchical evolutionary mining has been explored by [?], with complementary frameworks such as AlphaForge [?], Alphanetv4 [?], and Evolutionary Alpha [?]. Recent LLM-enhanced pipelines include AlphaGPT [?], AlphaGPT 2.0 [?], GPTSignal [?], QuantAgent [2], and REINFORCE-based factor mining [?], [?]. My approach differs by enforcing diversity *structurally* on a PCA 2D behavior space (grid-wise top-|IC| selection), avoiding full pairwise correlation scans while scaling search across depths and generations.

3 Methodology

I outline data handling, the information coefficient (IC), the GA + PCA–grid QD pipeline, and the DNN-based aggregation with XAI.

3.1 Data and Preprocessing

I use adjusted close (`adj_close`) to reflect dividends/splits. Missing data are handled with `ffill/bfill`. Data is normalized cross-sectionally using z-scores and winsorized daily.

3.2 Information Coefficient (IC)

For time t and universe $\mathcal{U}_t$, the alpha value of expression ϕ is $A_{i,t} = \phi(X_{i,t})$. I evaluate the *cross-sectional* Spearman rank correlation between $\{A_{\cdot,t}\}$ and next-day returns $\{r_{\cdot,t+1}\}$:

$$\text{IC}_t(\phi) = \rho\Big(\text{rank}(A_{\cdot,t}),\ \text{rank}(r_{\cdot,t+1})\Big), \qquad \overline{\text{IC}} = \tfrac{1}{T}\sum_t \text{IC}_t. \tag{1}$$

Industry interpretation: $|\text{IC}| > 0.02$ (significant), > 0.05 (good), > 0.10 (strong), > 0.15 (very strong).

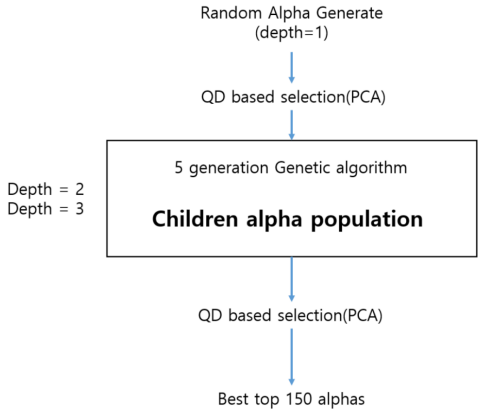


Fig. 2: Overall pipeline: hierarchical GA + PCA–grid QD selection.

3.3 Hierarchical GA with PCA–Grid QD Selection

3.3.0.1 Generation and evolution.: I initialize ∼3,000 depth-1 expressions from fields and operators (Appendix). Each generation: (i) evaluate IC, (ii) apply crossover/mutation under validity and max-depth checks, (iii) *select* parents by PCA–grid QD to maintain both quality and diversity, and (iv) repeat for 5 generations per depth before increasing maximum depth by one.

3.3.0.2 Behavior space and grid selection.: Vectorize each alpha's cross-sectional time series to $X \in \mathbb{R}^{p\times(nT)}$; project to $K = 2$ via PCA: $B = XW_2 \in \mathbb{R}^{p\times 2}$. On a $G \times G$ grid over the 2D plane, select per-cell the top candidates by |IC|. This *structurally* spreads selections across behavior space, avoiding correlation clustering with complexity $\approx O(p\,nT)$ (vs. $O(p^2nT)$ for pairwise scans).

3.4 DNN Aggregation and Portfolio Mapping

I train a DNN to predict $\hat{p}_{i,t} = \Pr(P_{i,t+5} > P_{i,t})$ from the top-150 alpha panel. The loss is binary cross-entropy (BCE) only:

$$\mathrm{BCE}(y,\hat{p}) = -\tfrac{1}{N}\sum_i \big(y_i \log \hat{p}_i + (1-y_i)\log(1-\hat{p}_i)\big). \quad (2)$$

Predictions are mapped to long-only portfolio weights via cross-sectional z-scores/ranks with $\sum_i w_{i,t} = 1$.

3.5 Explainable AI (XAI)

I apply SHAP to quantify each alpha's contribution to the DNN output. Dominant signals (often liquidity/momentum blends) show consistently high positive SHAP; negative-IC alphas (mean-reversion/overheat) contribute on the sell side, improving diversity.

4 Experimental Results

I evaluate (i) a 3-name prototype (AAPL/MSFT/GOOG) and (ii) S&P 100. All ICs are daily next-day Spearman; the DNN label is 5-day up/down.

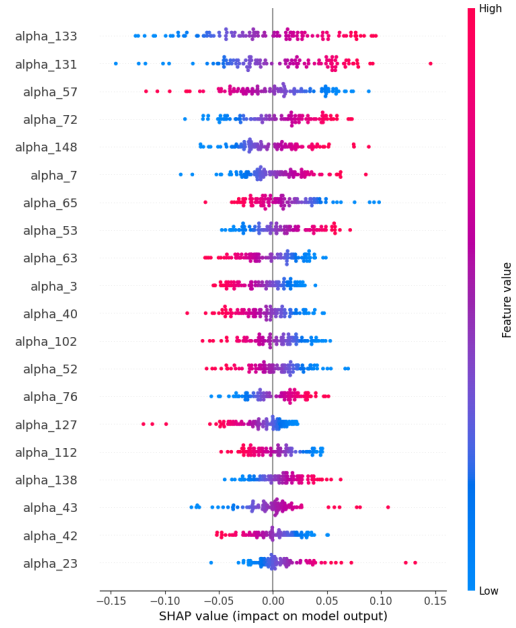


Fig. 3: SHAP summary: contribution of individual alphas to the Mega Alpha.

4.1 Selection by QD Scoring

Per generation I merge parents/children, compute ICs, embed to PCA(2), and select per grid cell the top-|IC| candidates, maintaining a 150-size parent pool for the next round.

4.2 Key Outcomes (S&P 100)

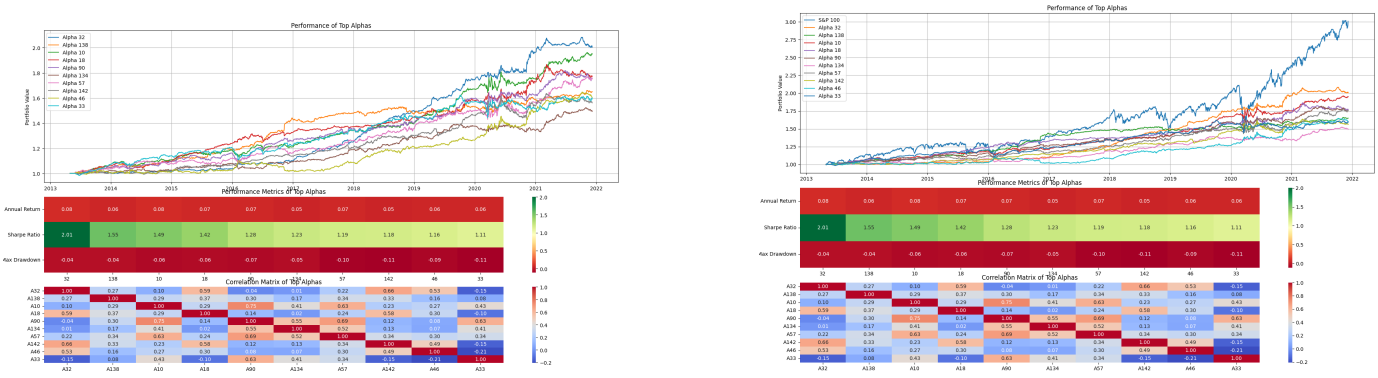
- **Average IC:** $\sim 0.001 \rightarrow \sim 0.004$ ($\sim 4\times$).
- **Significant alphas:** Many satisfy $|\mathrm{IC}| > 0.02$.
- **Portfolio (test):** $\sim$ 60% cumulative return; rolling Sharpe/Sortino ≥ 1.
- **Risk:** Max drawdown $\sim 40\%$ due to long-only exposure.

5 Discussion

The PCA-grid approach provides computational advantages over traditional correlation-based diversity measures. Rather than computing pairwise correlations for thousands of candidates, the method projects alpha behaviors into a 2D space and enforces diversity structurally through grid-based selection. This scales better as the candidate pool grows and prevents clustering around similar expression patterns.

The choice of binary cross-entropy loss, while simple, isolates the aggregation benefit from factor-specific metrics. More sophisticated loss functions incorporating IC or Sharpe ratio terms could potentially improve performance but would confound the contribution of the neural network combination versus the underlying factor quality.

Several limitations affect the current approach. The long-only constraint produces significant drawdowns during market downturns, as evidenced by the 40% maximum drawdown. The framework also lacks transaction cost modeling and does not account for alpha decay over time. Additionally, the 5-day prediction horizon may not capture longer-term market dynamics that could benefit portfolio construction.



(a) Alpha backtest results

(b) Alpha backtest with S&P 100

Fig. 4: Alpha backtests (illustrative). The GA + QD pipeline supplies a robust pool for aggregation.

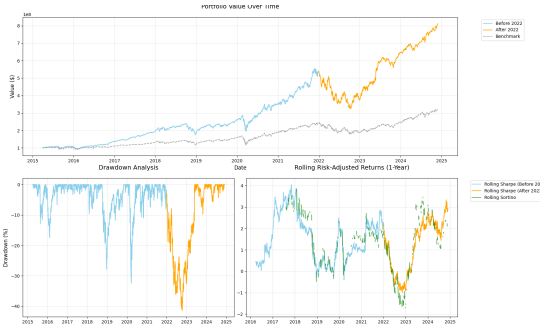


Fig. 5: Mega Alpha (DNN-aggregated) portfolio; rolling Sharpe/Sortino ≥ 1 on test.

6 Conclusion

This work demonstrates that systematic alpha factor mining can benefit from explicit diversity control during the generation process. The hierarchical genetic algorithm with PCA-grid quality-diversity selection produces factors with improved information coefficients compared to random generation, while the neural network aggregation provides additional performance gains over individual factors.

The S&P 100 results show a four-fold improvement in average IC and 60% cumulative returns with reasonable risk-adjusted metrics, though the long-only approach limits downside protection. The framework scales computationally and maintains interpretability through SHAP analysis, making it suitable for practical implementation.

Future extensions should incorporate long-short capabilities, transaction cost modeling, and adaptive factor lifecycle management to address the current limitations while maintaining the core benefits of systematic generation and diversity-aware selection.

Appendix A
Definitions of Data Fields and Operator Functions

This appendix details the data fields and operator functions used in alpha generation. Each function is expressed here in mathematical form.

A.1 Data Fields

Let P_t denote the price at time t and V_t the trading volume. The basic data fields are defined as:

- `OPEN`: Opening price, O_t.
- `HIGH`: Highest price, H_t.
- `LOW`: Lowest price, L_t.
- `CLOSE`: Adjusted Closing price, C_t.
- `VOLUME`: Trading volume, V_t.

Additional fields such as `ADV20` (20-day average daily volume) are computed as:

$$\texttt{ADV20}(t) = \frac{1}{20}\sum_{i=0}^{19} V_{t-i}.$$

A.2 Operator Functions

I define the following functions for constructing alpha expressions:

Arithmetic Operators

$$\begin{aligned}
\text{safe_add}(a,b) &= a+b,\\
\text{safe_subtract}(a,b) &= a-b,\\
\text{safe_multiply}(a,b) &= a\times b,\\
\text{safe_divide}(a,b) &= \frac{a}{b+\epsilon}, \quad \epsilon>0.
\end{aligned}$$

Transformation Functions

$$\begin{aligned}
\text{normalize}(x) &= \frac{x-\mu(x)}{\sigma(x)+\epsilon},\\
\text{scale}(x) &= \frac{x-\min(x)}{\max(x)-\min(x)+\epsilon},\\
\text{abs}(x) &= |x|.
\end{aligned}$$

Here, $\mu(x)$ and $\sigma(x)$ denote the mean and standard deviation of x, respectively.

Time-Series Functions

$$\begin{aligned}
\text{ts_delay}(x, n) &= x_{t-n}, \\
\text{ts_delta}(x, n) &= x_t - x_{t-n}, \\
\text{ts_returns}(x, n) &= \frac{x_t - x_{t-n}}{x_{t-n} + \epsilon}, \\
\text{ts_mean}(x, n) &= \frac{1}{n} \sum_{i=0}^{n-1} x_{t-i}, \\
\text{ts_std}(x, n) &= \sqrt{\frac{1}{n} \sum_{i=0}^{n-1} \left(x_{t-i} - \text{ts_mean}(x, n)\right)^2}.
\end{aligned}$$

Cross-Sectional Functions

For a vector $x = [x_1, x_2, \ldots, x_N]$:

$$\begin{aligned}
\text{cs_rank}(x) &= \text{rank}(x), \\
\text{scale}_{\text{cs}}(x_i) &= \frac{x_i}{\sum_{j=1}^{N} |x_j| + \epsilon}, \\
\text{normalize}_{\text{cs}}(x) &= \frac{x - \mu(x)}{\sigma(x) + \epsilon}.
\end{aligned}$$

Note: ϵ is a small positive constant to prevent division by zero and ensure numerical stability.

REFERENCES

[1] Z. Kakushadze, "101 Formulaic Alphas," *arXiv preprint arXiv:1601.00991*, 2016. [Online]. Available: https://arxiv.org/abs/1601.00991

[2] S. Wang, H. Yuan, L. M. Ni, and J. Guo, "QuantAgent: Seeking Holy Grail in Trading by Self-Improving Large Language Model," *arXiv preprint arXiv:2402.03755*, 2024. [Online]. Available: https://arxiv.org/abs/2402.03755

[3] Y. Li *et al.*, "Chain-of-Alpha: Unleashing the Power of Large Language Models for Alpha Mining in Quantitative Trading," *arXiv preprint arXiv:2508.06312*, 2024. [Online]. Available: https://arxiv.org/abs/2508.06312

[4] X. Chen *et al.*, "Automate Strategy Finding with LLM in Quant Investment," *arXiv preprint arXiv:2409.06289*, 2024. [Online]. Available: https://arxiv.org/abs/2409.06289

[5] K. Xiao *et al.*, "TradingAgents: Multi-Agents LLM Financial Trading Framework," *arXiv preprint arXiv:2412.20138*, 2024. [Online]. Available: https://arxiv.org/abs/2412.20138

[6] W. Chen, M. Jiang, W. G. Zhang, and Z. Chen, "AlphaEvolve: A Learning Framework to Discover Novel Alphas in Quantitative Investment," in *Proceedings of the 2021 International Conference on Management of Data*, 2021, pp. 1408–1416. [Online]. Available: https://dl.acm.org/doi/10.1145/3448016.3457324

[7] Y. Wang *et al.*, "Large Language Model Agent in Financial Trading: A Survey," *arXiv preprint arXiv:2408.06361*, 2024. [Online]. Available: https://arxiv.org/abs/2408.06361

[8] J. Li *et al.*, "From Deep Learning to LLMs: A survey of AI in Quantitative Investment," *arXiv preprint arXiv:2503.21422*, 2025. [Online]. Available: https://arxiv.org/abs/2503.21422